VENKATA THARUN PARSA

Agentic AI Engineer | RAG | Multi-Agent Systems | Cloud & MLOps

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PROFESSIONAL SUMMARY

Agentic AI Engineer specializing in the design and deployment of production-grade multi-agent systems, retrieval-augmented generation (RAG) pipelines, and vision-language models. Proven experience in architecting end-to-end AI systems integrating LLMs, multimodal models, and autonomous agents to solve real-world problems across compliance automation, cloud infrastructure management, and film pre-visualization. Experienced in designing scalable AI system architectures with containerized deployment, distributed processing workflows, real-time monitoring, and CI/CD-driven MLOps pipelines for production environments.

EDUCATION

B.Tech in Computer Science and Engineering — **CGPA: 8.5**

Rajiv Gandhi University of Knowledge Technologies (RGUKT), Basar — *Expected: 2027*

TECHNICAL SKILLS

Agentic AI & Generative AI: Multi-Agent Systems, RAG Pipelines, Prompt Engineering, LangChain, LangGraph, CrewAI, AutoGen

Multimodal & Vision Models: CLIP, BLIP, LoRA Fine-tuning, Vision-Language Models, Multimodal LLMs

Machine Learning & NLP: Semantic Search, Intent Classification, Embeddings, Vector Retrieval

Programming Languages: Python, Java

Cloud & MLOps: AWS (EC2, S3, IAM, CloudWatch), Azure (VM, VNet), Docker, Jenkins, CI/CD, FastAPI, Linux

PROJECTS

TaxSetu — Autonomous Multi-Agent GST Compliance Platform

Stack: LangChain, LangGraph, Gemini API, FAISS, FastAPI, AWS

- Architected a Master Planner system orchestrating 5 specialized AI agents for autonomous GST workflows.
- Engineered structured decision pipelines enabling explainable and audit-ready AI outputs.
- Integrated OCR, tax computation, and reasoning pipelines for end-to-end automation.
- Reduced MSME compliance complexity through fully autonomous execution.
- Designed for 7.34 crore MSMEs with multilingual accessibility.

VisionSync — Intelligent Character and Scene Planning System

Stack: Stability AI SDXL, LoRA, RAG, Gemini Vision, LangChain, FAISS

- Built a multimodal AI pipeline for film pre-visualization workflows.
- Achieved 95% character consistency across 100+ frames using LoRA fine-tuning.
- Implemented RAG-based validation for visual consistency.
- Designed a 6-agent system for scene planning and production risk prediction.

- Reduced storyboard generation time by 90% and API usage by 80%.

NINA — Voice-Driven AI Browser Automation Agent

Stack: Playwright, Python, NLP, FastAPI, DOM Parsing

- Built a voice-to-action pipeline enabling real-time browser automation.
- Designed dynamic DOM understanding using a Field Registry system.
- Enabled plug-and-play SDK integration with zero manual configuration.
- Implemented robust multi-step task execution with session recovery.

AstraDeploy — Scalable AI Deployment and Monitoring Platform

Stack: AWS, Docker, Jenkins, Grafana, CI/CD

- Built a cloud-native MLOps platform for deploying AI systems at scale.
- Designed automated CI/CD pipelines that significantly reduced deployment time.
- Implemented containerized and auto-scalable infrastructure.
- Integrated real-time monitoring and observability via Grafana and CloudWatch.

InfraGenie — Agentic Cloud Infrastructure Platform

Stack: Python, FastAPI, Gemini API, ChromaDB, RAG, Terraform, AWS, Docker, React

- Built a production-grade multi-agent AI system that provisions, monitors, and autonomously manages AWS infrastructure from plain English instructions.
- Architected a 5-agent system (Orchestrator, Planner, Executor, Monitor, Security) with a 4-tier intelligence router — hardcoded rules, RAG lookup, Gemini reasoning, and human-in-the-loop gates.
- Engineered a self-improving RAG pipeline using ChromaDB and Gemini embeddings across 6 knowledge collections including deployment history, incident logs, and AWS best practices.
- Integrated Terraform CLI, tfsec, Checkov, and Infracost for automated infrastructure generation, security scanning, and cost governance before every deployment.
- Implemented human-in-the-loop control for irreversible actions with autonomous self-correction loops and real-time WebSocket streaming to a React dashboard.

HACKATHON ACHIEVEMENTS

- Winner — Smart India Hackathon (Internal), RGUKT Basar | 2025
- Top 6 Finalist — Cine AI Hackathon, T-Works Hyderabad | Jan 2026
- Participant — Build for India Hackathon, Kerala Startup Mission | Feb 2026
- Participant — Tutedude Hackathon | 2025

EXPERIENCE

Azure Cloud Intern

Microsoft Elevate Program

Microsoft India

- Deployed and managed cloud infrastructure including VMs, storage, and networking.
- Applied monitoring and scalability practices relevant to AI system deployment.
- Supported AI deployment workflows using cloud and MLOps principles.

CERTIFICATIONS

- IBM RAG and Agentic AI Professional Certificate
- IBM DevOps and Software Engineering Professional Certificate